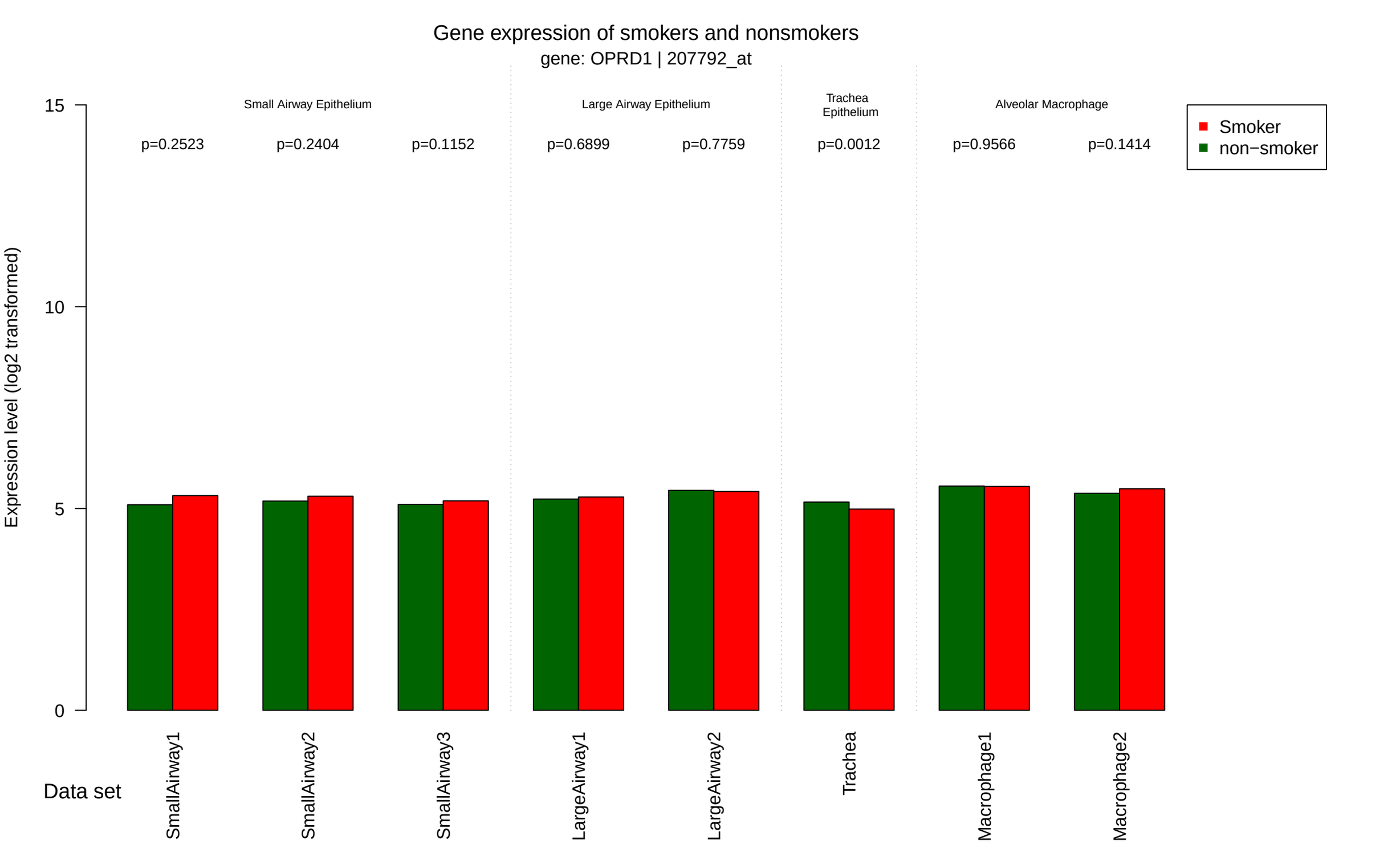
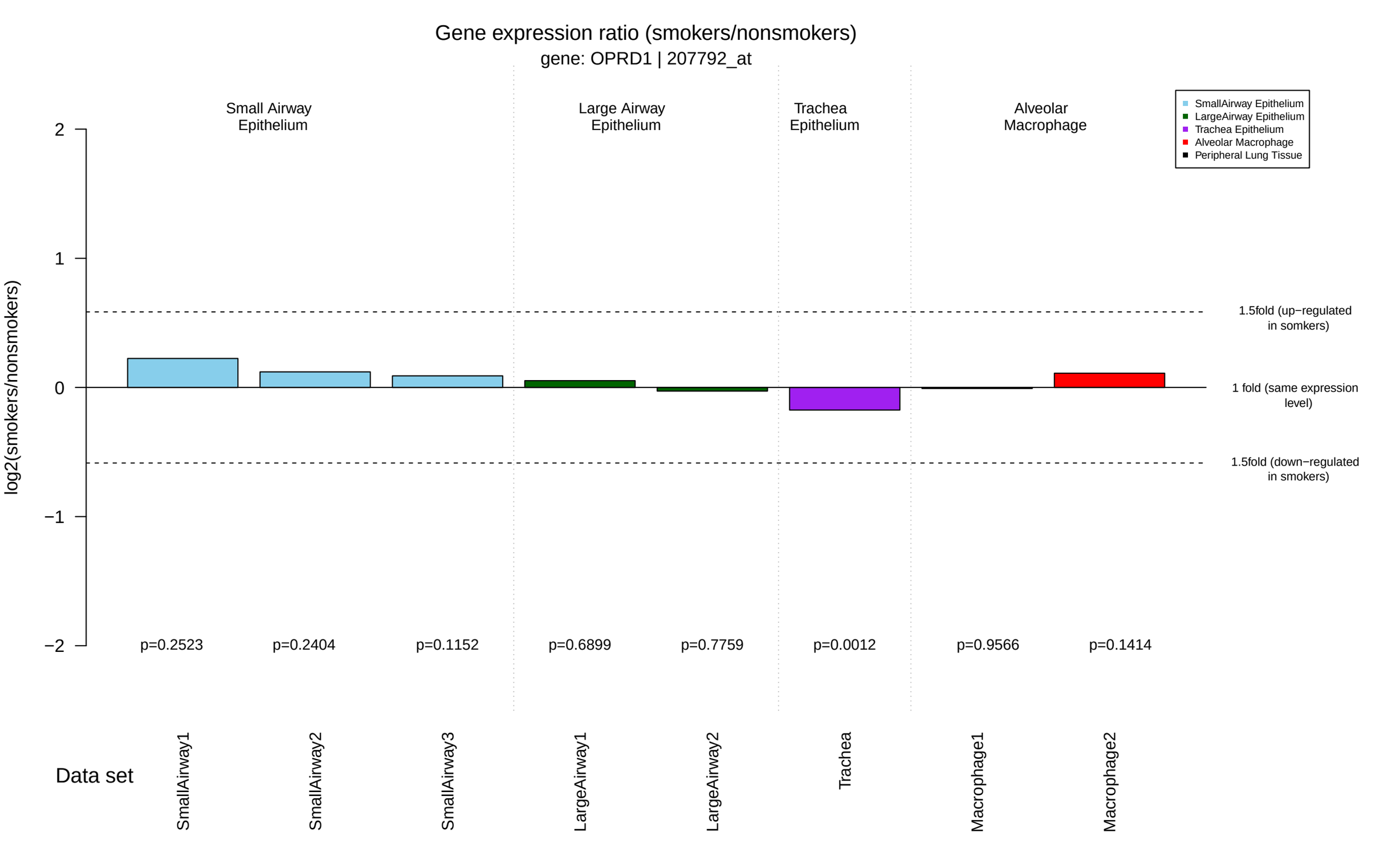
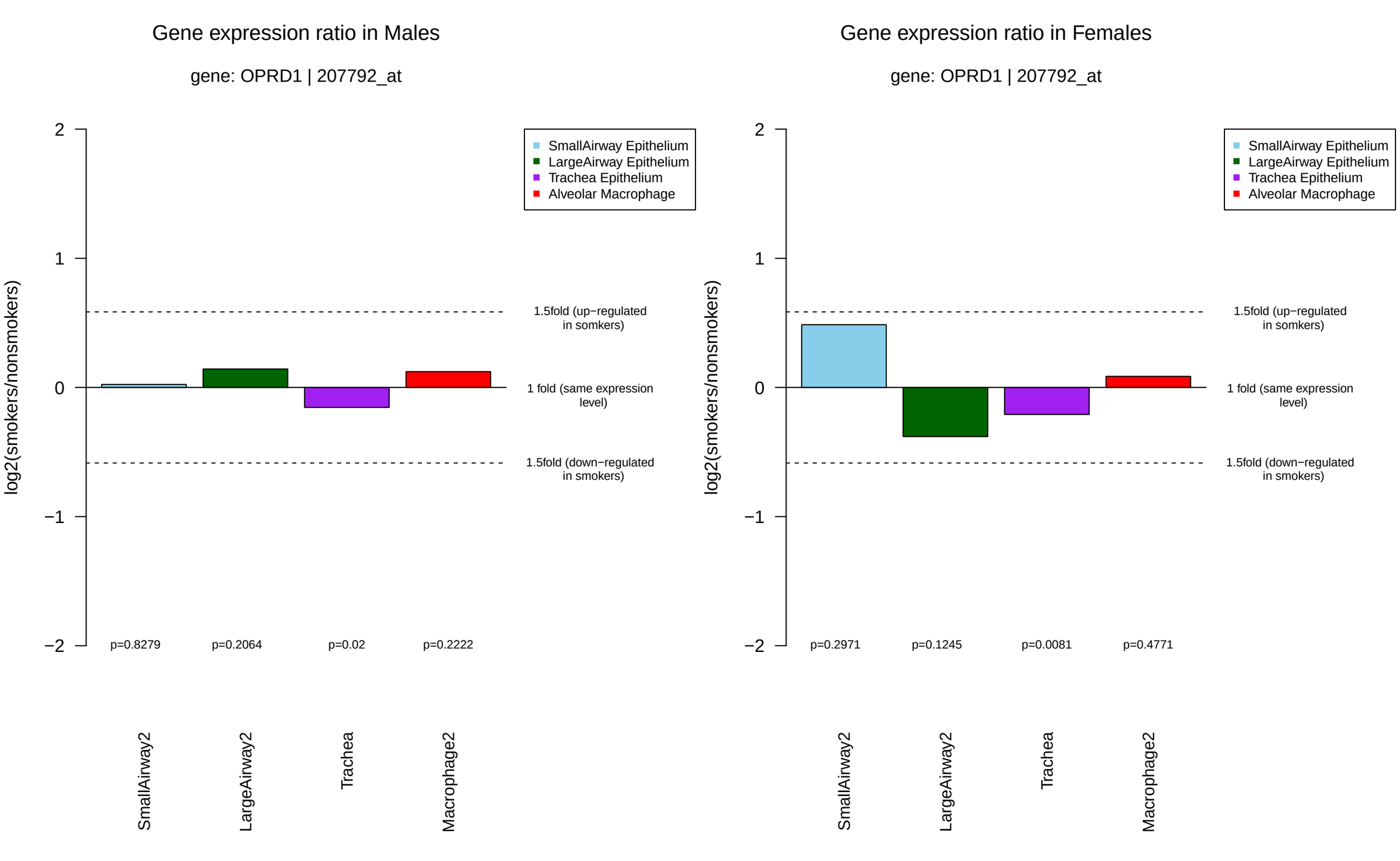
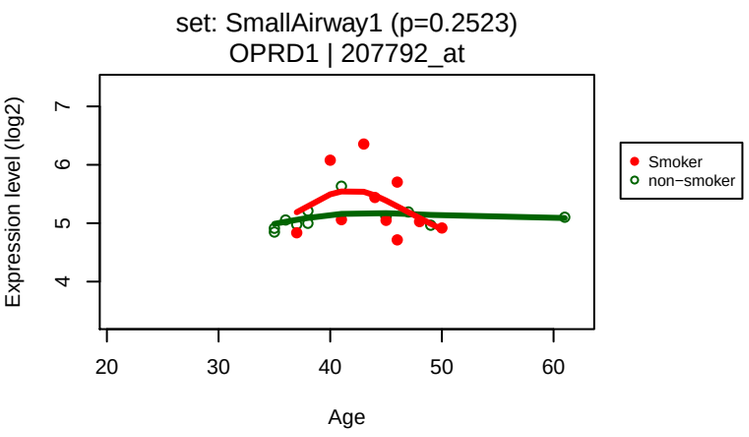
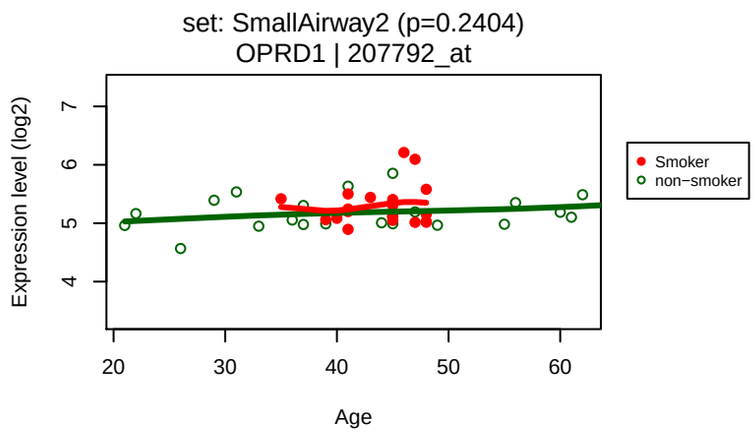
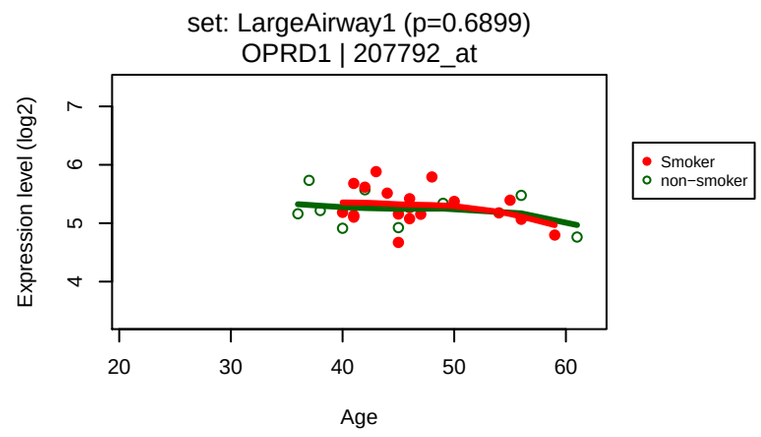

Gene expression of smokers and nonsmokers
gene: OPRD1 | 207792_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2523
p=0.2404
p=0.1152
p=0.6899
p=0.7759
p=0.0012
p=0.9566
p=0.1414
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

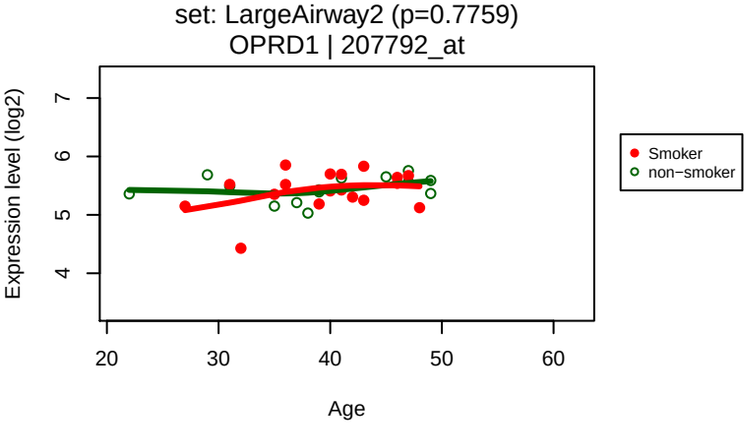

Gene expression ratio (smokers/nonsmokers)
gene: OPRD1 | 207792_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2523
p=0.2404
p=0.1152
p=0.6899
p=0.7759
p=0.0012
p=0.9566
p=0.1414
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

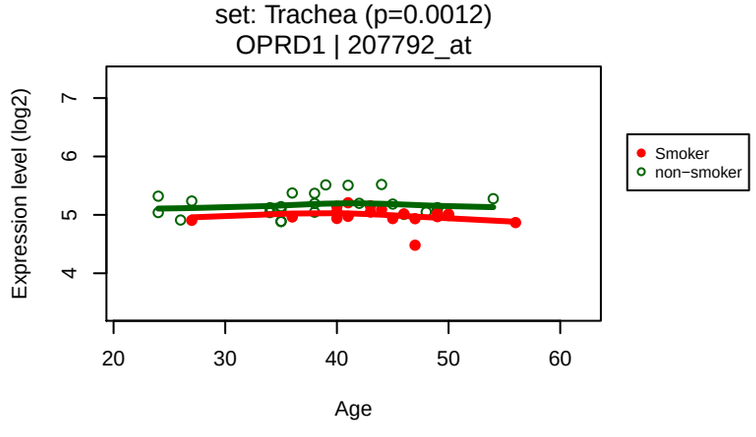

Gene expression ratio in Males
gene: OPRD1 | 207792_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8279
p=0.2064
p=0.02
p=0.2222
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: OPRD1 | 207792_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2971
p=0.1245
p=0.0081
p=0.4771
SmallAirway2
LargeAirway2
Trachea
Macrophage2

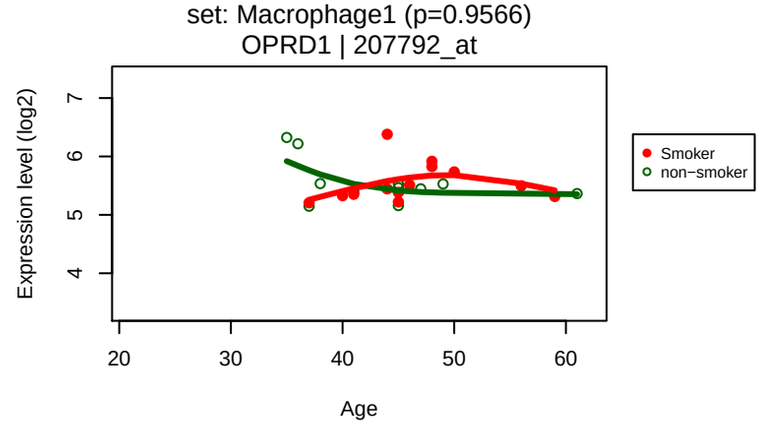

set: SmallAirway1 (p=0.2523)
OPRD1 | 207792_at
Expression level (log2)
Age
Smoker
non−smoker

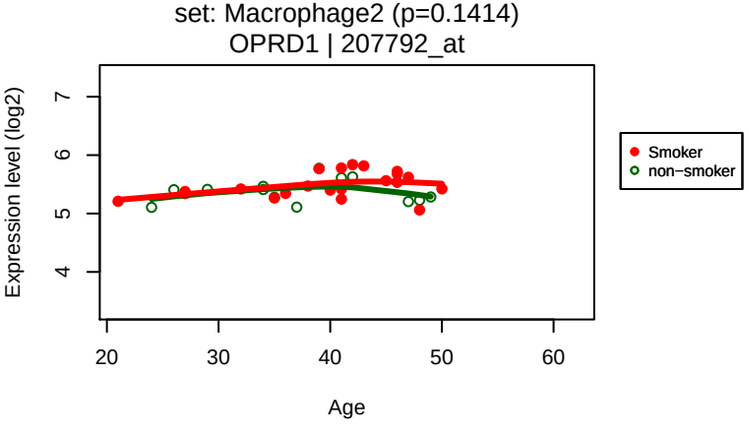

set: SmallAirway2 (p=0.2404)
OPRD1 | 207792_at
Expression level (log2)
Age
Smoker
non−smoker

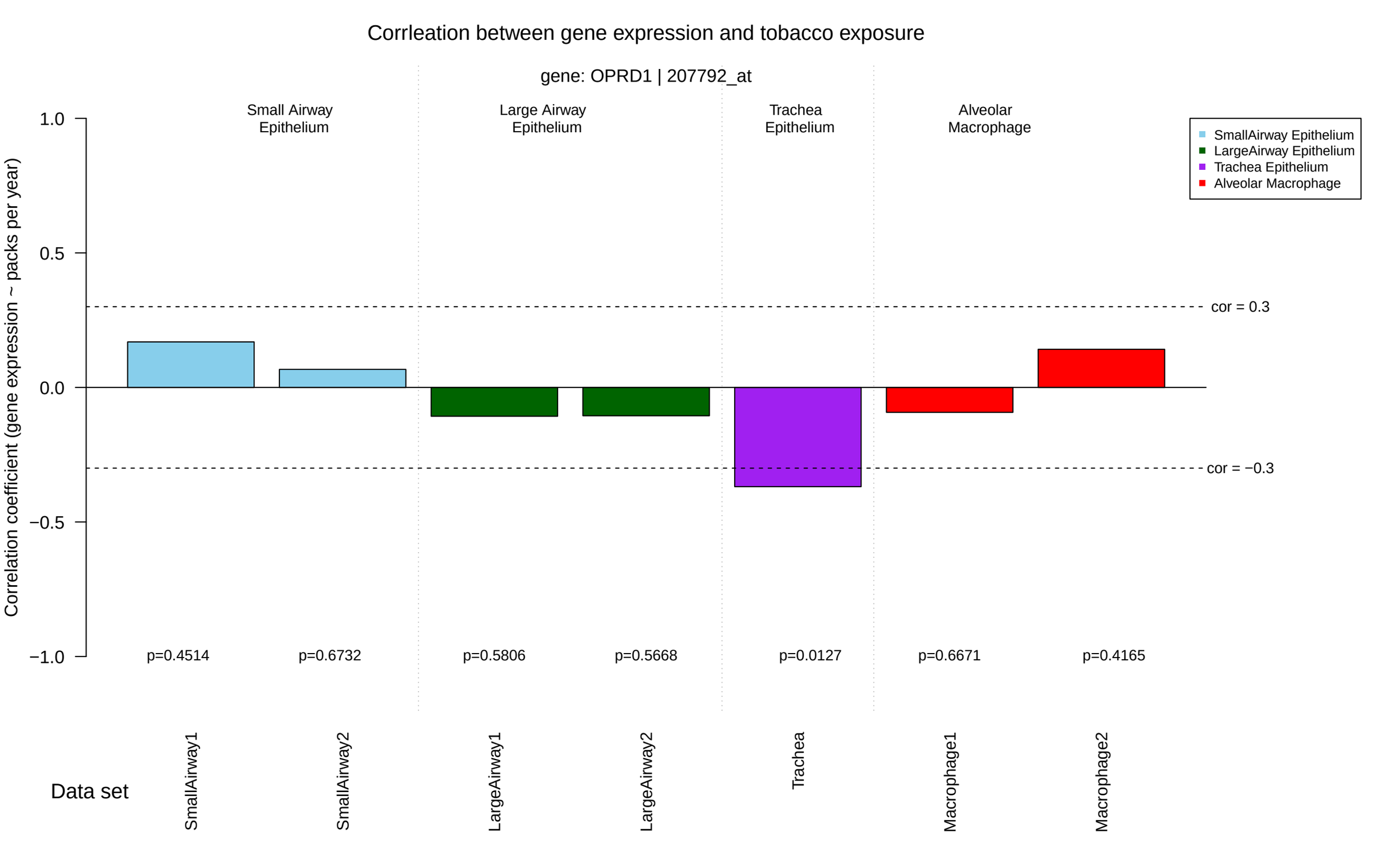

set: LargeAirway1 (p=0.6899)
OPRD1 | 207792_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7759)
OPRD1 | 207792_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0012)
OPRD1 | 207792_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9566)
OPRD1 | 207792_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1414)
OPRD1 | 207792_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: OPRD1 | 207792_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4514
p=0.6732
p=0.5806
p=0.5668
p=0.0127
p=0.6671
p=0.4165
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2